Mathematical Methods

Module 4 – Exam Exercise Pack

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Stochastic processes, filtrations, martingales, stopping times, Markov dynamics, Brownian motion and Poisson processes

How to use this exercise pack

- Each exercise is designed as a possible written-exam task.
- Every solution is written step by step, with the object, the formula and the conclusion made explicit.
- Difficulty is marked from $\star$ to $\star\star\star\star\star$.
- The exercises cover the whole Module 4 programme: processes, laws, stationarity, filtrations, martingales, stopping times, Markov chains, Brownian motion and Poisson processes.

Intuition: Study method

Before looking at the solution, identify the mathematical object: random variable, event, process, filtration, martingale, stopping time, transition matrix, Brownian motion or Poisson process.

Index of the exercise pack

- Processes, paths and finite-dimensional laws.
- Stationarity, increments and autocovariance.
- Filtrations, adaptation and predictability.
- Conditional expectation and martingale checks.
- Martingales, submartingales and transforms.
- Stopping times and optional sampling.
- Discrete-time Markov chains.
- Continuous-time Markov dynamics.
- Brownian motion.
- Poisson processes and counting models.

Module 4 Exercise Pack

Processes, paths and finite-dimensional laws

Exercise 1 – Objects in a process

Exam-style exercise: Objects in a process ⋆

Let $(X_t)_{t\geq 0}$ be a stochastic process. Explain the difference between X_3, $X_3(\omega)$ and the map $t \mapsto X_t(\omega)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 1 – Objects in a process

Solution: Detailed solution

1. X_3 is a random variable. Before observing the outcome, it has a distribution.
2. $X_3(\omega)$ is a real number: the value of X_3 for the fixed outcome ω.
3. The map $t \mapsto X_t(\omega)$ is one entire trajectory of the process.
4. Thus the same notation describes three levels: random variable, realized number and path.

Exercise 2 – Build a partial-sum process

Exam-style exercise: Build a partial-sum process ⋆

Let $\xi_1, \xi_2, \ldots$ be random variables and define $S_0 = 0$, $S_n = \sum_{k=1}^{n} \xi_k$. Is $(S_n)_{n \geq 0}$ a stochastic process? What is its state space if each $\xi_k \in \{-2, 1\}$?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 2 – Build a partial-sum process

Solution: Detailed solution

1. A stochastic process is a family of random variables indexed by time or step number.
2. For each n, S_n is a random variable because it is a finite sum of random variables.
3. Therefore $(S_n)_{n\geq 0}$ is a discrete-time stochastic process.
4. Since each increment is an integer, S_n takes integer values. The natural state space is $\mathbb{Z}$, although not every integer is necessarily reachable at a fixed n.

Exercise 3 – Same marginals, different processes

Exam-style exercise: Same marginals, different processes ⋆⋆

Let $Z \sim N(0,1)$ and set $X_t = Z$ for every integer $t \geq 0$. Let (Y_t) be independent $N(0,1)$ random variables. Do the two processes have the same marginal distributions? Do they have the same process law?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 3 – Same marginals, different processes

Solution: Detailed solution

1. For every t, $X_t = Z \sim N(0,1)$.
2. For every t, $Y_t \sim N(0,1)$ by construction.
3. Hence the one-time marginal distributions coincide.
4. But the joint laws differ: $X_0 = X_1$ almost surely, while Y_0 and Y_1 are independent and are not equal almost surely.
5. Therefore equal marginals do not imply equal process laws.

Exercise 4 – Finite-dimensional distribution

Exam-style exercise: Finite-dimensional distribution ⋆⋆

Let $X_n = (-1)^n Z$, where Z has mean zero and variance one. Find the law of the vector (X_0, X_1, X_2) in terms of Z.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 4 – Finite-dimensional distribution

Solution: Detailed solution

1. Substitute the definition at each time:

$$X_0 = Z, \qquad X_1 = -Z, \qquad X_2 = Z.$$

2. Therefore

$$(X_0, X_1, X_2) = (Z, -Z, Z).$$

3. The finite-dimensional distribution is the distribution of this vector.
4. It is concentrated on the line $\{(z, -z, z) : z \in \mathbb{R}\}$, not on all of $\mathbb{R}^3$.

Exercise 5 – Crossing event

Exam-style exercise: Crossing event ⋆

For a discrete-time process $(X_k)_{k\geq 0}$, write the event "the process crosses the level a before or at time n" in two equivalent ways.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 5 – Crossing event

Solution: Detailed solution

1. Crossing the level means that at least one of the values $X_0, \ldots, X_n$ is at least a.
2. Hence the event is
$$\left\{ \max_{0 \le k \le n} X_k \ge a \right\}.$$
3. Equivalently, it is the union of the single-time events:
$$\bigcup_{k=0}^{n} \{X_k \ge a\}.$$
4. The union form is often useful for proving measurability.

Exercise 6 – First hitting time as a random variable

Exam-style exercise: First hitting time as a random variable ⋆⋆

Let $\tau_a = \inf\{n \geq 0 : X_n \geq a\}$ for a discrete-time process. Express the event $\{\tau_a \leq n\}$ using $X_0, \ldots, X_n$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 6 – First hitting time as a random variable

Solution: Detailed solution

1. The event $\{\tau_a \leq n\}$ means that the first crossing occurs no later than time n.
2. This is equivalent to saying that some value among $X_0, \ldots, X_n$ is at least a.
3. Therefore

$$\{\tau_a \leq n\} = \bigcup_{k=0}^{n} \{X_k \geq a\} = \left\{ \max_{0 \leq k \leq n} X_k \geq a \right\}.$$

Exercise 7 – Random walk distribution

Exam-style exercise: Random walk distribution ⋆⋆

Let $S_n = \sum_{k=1}^{n} \xi_k$, where $\mathbb{P}(\xi_k = 1) = p$ and $\mathbb{P}(\xi_k = -1) = 1 - p$. Find $\mathbb{P}(S_n = s)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 7 – Random walk distribution

Solution: Detailed solution

1. Let K_n be the number of $+1$ increments. Then $K_n \sim \mathrm{Bin}(n, p)$.
2. If $K_n = k$, then $S_n = k - (n - k) = 2k - n$.
3. Thus $S_n = s$ means $k = (n + s)/2$.
4. Therefore
$$\mathbb{P}(S_n = s) = \binom{n}{(n+s)/2} p^{(n+s)/2} (1-p)^{(n-s)/2},$$
provided $(n + s)/2$ is an integer between 0 and n; otherwise the probability is zero.

Exercise 8 – Mean and variance of random walk

Exam-style exercise: Mean and variance of random walk ⋆⋆

For the same random walk, compute $\mathbb{E}[S_n]$ and $\mathrm{Var}(S_n)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 8 – Mean and variance of random walk

Solution: Detailed solution

1. For one increment,
$$\mathbb{E}[\xi_k] = p - (1-p) = 2p - 1.$$
2. By linearity,
$$\mathbb{E}[S_n] = \sum_{k=1}^{n} \mathbb{E}[\xi_k] = n(2p-1).$$
3. Since $\xi_k^2 = 1$, $\mathrm{Var}(\xi_k) = 1 - (2p-1)^2 = 4p(1-p)$.
4. Independence gives
$$\mathrm{Var}(S_n) = \sum_{k=1}^{n} \mathrm{Var}(\xi_k) = 4np(1-p).$$

Exercise 9 – Law versus path statement

Exam-style exercise: Law versus path statement ⋆⋆

Decide whether the statement is distributional or pathwise: (i) $X_t \sim N(0,t)$; (ii) $t \mapsto X_t(\omega)$ is continuous; (iii) $X_t - X_s \sim N(0, t-s)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 9 – Law versus path statement

Solution: Detailed solution

1. $X_t \sim N(0,t)$ is a distributional statement about one random variable.
2. $t \mapsto X_t(\omega)$ is continuous is a pathwise statement after fixing the outcome.
3. $X_t - X_s \sim N(0, t-s)$ is distributional because it describes the law of an increment.
4. In process theory, confusing distributional and pathwise statements is a common error.

Exercise 10 – State-space classification

Exam-style exercise: State-space classification ⋆

Give a natural state space for each process: a simple random walk, a two-dimensional random walk, a Poisson process, and a finite-state weather chain with states sunny/rainy/cloudy.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 10 – State-space classification

Solution: Detailed solution

1. Simple random walk: usually $\mathbb{Z}$.
2. Two-dimensional random walk: $\mathbb{Z}^2$.
3. Poisson process: $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$.
4. Weather chain: the finite set $\{\text{sunny}, \text{rainy}, \text{cloudy}\}$.

Module 4 Exercise Pack

Stationarity, increments and autocovariance

Exercise 11 – Check weak stationarity

Exam-style exercise: Check weak stationarity ⋆⋆

Let $X_n = Z + \varepsilon_n$, where $Z, \varepsilon_1, \varepsilon_2, \ldots$ are independent, $\mathbb{E}[Z] = \mathbb{E}[\varepsilon_n] = 0$, $\mathrm{Var}(Z) = a^2$, $\mathrm{Var}(\varepsilon_n) = \sigma^2$. Is (X_n) weakly stationary? Compute $\gamma(h)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 11 – Check weak stationarity

Solution: Detailed solution

1. Mean: $\mathbb{E}[X_n] = \mathbb{E}[Z] + \mathbb{E}[\varepsilon_n] = 0$, constant in n.
2. Variance: $\mathrm{Var}(X_n) = \mathrm{Var}(Z) + \mathrm{Var}(\varepsilon_n) = a^2 + \sigma^2$, constant in n.
3. For $h \neq 0$,
$$\mathrm{Cov}(X_{n+h}, X_n) = \mathrm{Cov}(Z + \varepsilon_{n+h}, Z + \varepsilon_n) = \mathrm{Var}(Z) = a^2.$$
4. Also $\gamma(0) = a^2 + \sigma^2$.
5. Thus the process is weakly stationary.

Exercise 12 – Trend and stationarity

Exam-style exercise: Trend and stationarity ⋆

Let $X_n = 3 + n + \varepsilon_n$, where (ε_n) is white noise with mean zero and variance σ^2. Is (X_n) weakly stationary?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 12 – Trend and stationarity

Solution: Detailed solution

1. Compute the mean:
$$\mathbb{E}[X_n] = 3 + n + \mathbb{E}[\varepsilon_n] = 3 + n.$$
2. This depends on n.
3. Weak stationarity requires a constant mean.
4. Therefore (X_n) is not weakly stationary, even though the noise part is stationary.

Exercise 13 – Alternating process

Exam-style exercise: Alternating process $\star\star\star$

Let $X_n = (-1)^n Z$, where $\mathbb{E}[Z] = 0$ and $\mathrm{Var}(Z) = 1$. Is (X_n) weakly stationary?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 13 – Alternating process

Solution: Detailed solution

1. Mean: $\mathbb{E}[X_n] = (-1)^n \mathbb{E}[Z] = 0$.
2. Variance: $\mathrm{Var}(X_n) = (-1)^{2n} \mathrm{Var}(Z) = 1$.
3. Autocovariance:

$$\mathrm{Cov}(X_{n+h}, X_n) = \mathbb{E}[(-1)^{n+h} Z \, (-1)^n Z] = (-1)^h \mathbb{E}[Z^2] = (-1)^h.$$

4. This depends only on h, not on n.
5. Hence the process is weakly stationary.

Exercise 14 – Random walk is not stationary

Exam-style exercise: Random walk is not stationary ⋆⋆

Let $S_n = \sum_{k=1}^{n} \xi_k$, with i.i.d. increments of mean zero and variance $\sigma^2 > 0$. Show that (S_n) is not weakly stationary.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 14 – Random walk is not stationary

Solution: Detailed solution

1. The mean is $\mathbb{E}[S_n] = 0$, so the mean condition is satisfied.
2. But
$$\mathrm{Var}(S_n) = \sum_{k=1}^{n} \mathrm{Var}(\xi_k) = n\sigma^2.$$
3. The variance depends on n and grows linearly.
4. Weak stationarity requires constant variance.
5. Therefore the random walk is not weakly stationary.

Exercise 15 – Stationary increments

Exam-style exercise: Stationary increments ⋆⋆

Let S_n be a random walk with i.i.d. increments. Show that $S_{n+k} - S_n$ has a distribution depending on k but not on n.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 15 – Stationary increments

Solution: Detailed solution

1. Write the increment:
$$S_{n+k} - S_n = \sum_{j=n+1}^{n+k} \xi_j.$$
2. The variables $\xi_{n+1}, \ldots, \xi_{n+k}$ are i.i.d. with the same law as $\xi_1, \ldots, \xi_k$.
3. Hence
$$S_{n+k} - S_n \overset{d}{=} \sum_{j=1}^{k} \xi_j = S_k.$$
4. Thus the distribution depends only on the length k.

Exercise 16 – Independent increments check

Exam-style exercise: Independent increments check ⋆⋆

Let $X_n = \xi_1 + \cdots + \xi_n$, where the ξ_k are independent. Show that $X_{t_1} - X_{t_0}, \ldots, X_{t_m} - X_{t_{m-1}}$ are independent for $0 \leq t_0 < t_1 < \cdots < t_m$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 16 – Independent increments check

Solution: Detailed solution

1. Each increment is a sum over a block of ξ variables:

$$X_{t_j} - X_{t_{j-1}} = \sum_{k=t_{j-1}+1}^{t_j} \xi_k.$$

2. The blocks of indices are disjoint.
3. Functions of independent disjoint families are independent.
4. Therefore the process has independent increments.

Exercise 17 – Autocorrelation of AR(1)

Exam-style exercise: Autocorrelation of AR(1) ⋆⋆⋆

Let $X_n = \rho X_{n-1} + \varepsilon_n$, where $|\rho| < 1$, $\mathbb{E}[\varepsilon_n] = 0$, $\mathrm{Var}(\varepsilon_n) = \sigma^2$, and the process is stationary. Find $\gamma(0)$ and $\gamma(h)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 17 – Autocorrelation of AR(1)

Solution: Detailed solution

1. From stationarity,
$$\mathrm{Var}(X_n) = \rho^2 \,\mathrm{Var}(X_{n-1}) + \sigma^2.$$
2. Thus $\gamma(0) = \sigma^2/(1-\rho^2)$.
3. For $h \geq 1$,
$$\gamma(h) = \mathrm{Cov}(X_{n+h}, X_n) = \rho \,\mathrm{Cov}(X_{n+h-1}, X_n) = \rho\gamma(h-1).$$
4. Therefore
$$\gamma(h) = \rho^h \gamma(0) = \frac{\sigma^2 \rho^h}{1-\rho^2}.$$

Exercise 18 – White noise covariance

Exam-style exercise: White noise covariance ⋆

Let (ε_n) be white noise with mean 0 and variance σ^2. Compute $\gamma(h) = \mathrm{Cov}(\varepsilon_{n+h}, \varepsilon_n)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 18 – White noise covariance

Solution: Detailed solution

1. For $h = 0$, $\gamma(0) = \mathrm{Var}(\varepsilon_n) = \sigma^2$.
2. For $h \neq 0$, white noise is uncorrelated across different times.
3. Hence $\gamma(h) = 0$ for $h \neq 0$.
4. Therefore

$$\gamma(h) = \begin{cases} \sigma^2, & h = 0, \\ 0, & h \neq 0. \end{cases}$$

Exercise 19 – Uncorrelated is not independent

Exam-style exercise: Uncorrelated is not independent ⋆⋆

Let $U \sim \mathrm{Unif}(-1, 1)$, $X = U$ and $Y = U^2$. Show that X and Y are uncorrelated but not independent.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 19 – Uncorrelated is not independent

Solution: Detailed solution

1. Since the distribution of U is symmetric, $\mathbb{E}[U] = 0$ and $\mathbb{E}[U^3] = 0$.
2. Thus
$$\mathrm{Cov}(X, Y) = \mathbb{E}[U^3] - \mathbb{E}[U]\mathbb{E}[U^2] = 0.$$
3. However, Y is completely determined by X: $Y = X^2$.
4. For example, knowing $X = 1/2$ implies $Y = 1/4$.
5. Hence X and Y are not independent.

Exercise 20 – Strict stationarity of i.i.d. sequence

Exam-style exercise: Strict stationarity of i.i.d. sequence $\star\star$

Prove that an i.i.d. sequence $(\xi_n)_{n\geq 0}$ is strictly stationary.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 20 – Strict stationarity of i.i.d. sequence

Solution: Detailed solution

1. Strict stationarity requires equality in law of

$$(\xi_{t_1}, \ldots, \xi_{t_m}) \quad \text{and} \quad (\xi_{t_1+h}, \ldots, \xi_{t_m+h}).$$

2. Since the variables are i.i.d., both vectors have product distribution $\mu^{\otimes m}$, where μ is the common law.
3. Therefore all finite-dimensional distributions are invariant under shifts.
4. Hence the sequence is strictly stationary.

Exercise 21 – Gaussian exception

Exam-style exercise: Gaussian exception ⋆⋆⋆

Explain why, for a Gaussian process, weak stationarity implies strict stationarity.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 21 – Gaussian exception

Solution: Detailed solution

1. A Gaussian finite-dimensional distribution is completely determined by its mean vector and covariance matrix.
2. Weak stationarity gives a constant mean and covariance depending only on time differences.
3. Therefore the mean vector of $(X_{t_1}, \ldots, X_{t_m})$ equals that of the shifted vector.
4. The covariance matrices are also equal after shifting all times by the same amount.
5. Hence the finite-dimensional distributions coincide: the Gaussian process is strictly stationary.

Module 4 Exercise Pack

Filtrations, adaptation and predictability

Exercise 22 – Natural filtration

Exam-style exercise: Natural filtration $\star$

For a discrete-time process (X_n), define its natural filtration. Interpret it in words.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 22 – Natural filtration

Solution: Detailed solution

1. The natural filtration is
$$\mathcal{F}_n^X = \sigma(X_0, X_1, \ldots, X_n).$$
2. It contains all events whose truth can be decided by observing the process up to time n.
3. In words: it is the information generated by the history of the process up to time n.

Exercise 23 – Adapted process

Exam-style exercise: Adapted process ⋆

Let $X_n = \xi_1 + \cdots + \xi_n$ and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$. Is (X_n) adapted to $(\mathcal{F}_n)$?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 23 – Adapted process

Solution: Detailed solution

1. A process is adapted if X_n is $\mathcal{F}_n$-measurable for every n.
2. Here X_n is a function of $\xi_1, \ldots, \xi_n$.
3. Since $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$, every measurable function of these variables is $\mathcal{F}_n$-measurable.
4. Therefore (X_n) is adapted.

Exercise 24 – Not adapted

Exam-style exercise: Not adapted ⋆⋆

Let $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$ and define $Y_n = \xi_{n+1}$. Is (Y_n) adapted to $(\mathcal{F}_n)$?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 24 – Not adapted

Solution: Detailed solution

1. For adaptation, Y_n must be $\mathcal{F}_n$-measurable.
2. But $Y_n = \xi_{n+1}$ depends on the next innovation.
3. The sigma-algebra $\mathcal{F}_n$ contains information only up to ξ_n.
4. Unless ξ_{n+1} is degenerate or already determined by the past, Y_n is not $\mathcal{F}_n$-measurable.
5. Thus (Y_n) is generally not adapted.

Exercise 25 – Predictability in discrete time

Exam-style exercise: Predictability in discrete time ⋆⋆

Let $H_n = X_{n-1}$ for $n \geq 1$, where (X_n) is adapted to $(\mathcal{F}_n)$. Is (H_n) predictable?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 25 – Predictability in discrete time

Solution: Detailed solution

1. In discrete time, H_n is predictable if H_n is $\mathcal{F}_{n-1}$-measurable.
2. Here $H_n = X_{n-1}$.
3. Since X is adapted, X_{n-1} is $\mathcal{F}_{n-1}$-measurable.
4. Therefore (H_n) is predictable.

Exercise 26 – Adapted but not predictable

Exam-style exercise: Adapted but not predictable ⋆⋆

Let $S_n = \sum_{k=1}^{n} \xi_k$ with natural filtration $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$. Is S_n predictable?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 26 – Adapted but not predictable

Solution: Detailed solution

1. S_n is adapted because it is known at time n.
2. Predictability would require S_n to be $\mathcal{F}_{n-1}$-measurable.
3. But
$$S_n = S_{n-1} + \xi_n,$$
and ξ_n is new information at time n.
4. Therefore S_n is generally not predictable.
5. The process S_{n-1} would be predictable, but S_n is not.

Exercise 27 – Measurability of an event

Exam-style exercise: Measurability of an event ⋆⋆

Let $\mathcal{F}_n = \sigma(X_0, \ldots, X_n)$. Show that the event $\{\max_{0 \le k \le n} X_k \le a\}$ belongs to $\mathcal{F}_n$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 27 – Measurability of an event

Solution: Detailed solution

1. Rewrite the event as
$$\left\{ \max_{0 \le k \le n} X_k \le a \right\} = \bigcap_{k=0}^{n} \{X_k \le a\}.$$
2. Since X_k is included in the generating variables of $\mathcal{F}_n$, each event $\{X_k \le a\}$ is in $\mathcal{F}_n$.
3. Sigma-algebras are closed under finite intersections.
4. Therefore the crossing-below event belongs to $\mathcal{F}_n$.

Exercise 28 – Information comparison

Exam-style exercise: Information comparison ⋆⋆

Suppose $\mathcal{G}_n \subseteq \mathcal{F}_n$ for every n. If a process is adapted to $(\mathcal{G}_n)$, is it adapted to $(\mathcal{F}_n)$?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 28 – Information comparison

Solution: Detailed solution

1. If the process is adapted to $(\mathcal{G}_n)$, then X_n is $\mathcal{G}_n$-measurable.
2. Since $\mathcal{G}_n \subseteq \mathcal{F}_n$, every $\mathcal{G}_n$-measurable random variable is also $\mathcal{F}_n$-measurable.
3. Hence X_n is $\mathcal{F}_n$-measurable.
4. Therefore adaptation to a smaller filtration implies adaptation to a larger filtration.

Exercise 29 – Future maximum is not current information

Exam-style exercise: Future maximum is not current information $\star\star$

Let $M_n = \max_{0\le k\le n} X_k$ and $N_n = \max_{0\le k\le n+1} X_k$. With $\mathcal{F}_n = \sigma(X_0,\dots,X_n)$, which of M_n, N_n is $\mathcal{F}_n$-measurable?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 29 – Future maximum is not current information

Solution: Detailed solution

1. M_n depends only on $X_0, \ldots, X_n$, so it is $\mathcal{F}_n$-measurable.
2. N_n also depends on X_{n+1}.
3. At time n, X_{n+1} is future information.
4. Therefore N_n is generally not $\mathcal{F}_n$-measurable.

Exercise 30 – Stopping-time measurability preview

Exam-style exercise: Stopping-time measurability preview ⋆⋆

Let $\tau = \inf\{n \geq 0 : X_n \geq a\}$. Show that $\{\tau = n\} \in \mathcal{F}_n$ for the natural filtration.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 30 – Stopping-time measurability preview

Solution: Detailed solution

1. The event $\{\tau = n\}$ means: no crossing before n and crossing at n.
2. Hence
$$\{\tau = n\} = \{X_0 < a, \dots, X_{n-1} < a\} \cap \{X_n \geq a\}.$$
3. Each event in this intersection depends only on $X_0, \dots, X_n$.
4. Therefore $\{\tau = n\} \in \mathcal{F}_n$.

Exercise 31 – Generated sigma-algebra

Exam-style exercise: Generated sigma-algebra $\star\star\star$

Let X be a random variable and let $Y = g(X)$ for a Borel function g. Show that $\sigma(Y) \subseteq \sigma(X)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 31 – Generated sigma-algebra

Solution: Detailed solution

1. To prove inclusion of sigma-algebras, it is enough to show that every event of the form $\{Y \in B\}$ is in $\sigma(X)$.
2. Since $Y = g(X)$,
$$\{Y \in B\} = \{g(X) \in B\} = \{X \in g^{-1}(B)\}.$$
3. Because g is Borel, $g^{-1}(B)$ is Borel whenever B is Borel.
4. Therefore $\{X \in g^{-1}(B)\} \in \sigma(X)$.
5. Hence $\sigma(Y) \subseteq \sigma(X)$.

Module 4 Exercise Pack

Conditional expectation and martingale checks

Exercise 32 – Conditional expectation on trivial information

Exam-style exercise: Conditional expectation on trivial information ⋆

Let X be integrable and let $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Compute $\mathbb{E}[X \mid \mathcal{F}_0]$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 32 – Conditional expectation on trivial information

Solution: Detailed solution

1. A $\mathcal{F}_0$-measurable random variable must be constant.
2. The conditional expectation must have the same integral as X over Ω.
3. Therefore the constant is $\mathbb{E}[X]$.
4. Hence

$$\mathbb{E}[X \mid \mathcal{F}_0] = \mathbb{E}[X].$$

Exercise 33 – Conditioning on full information

Exam-style exercise: Conditioning on full information $\star$

Let X be integrable and $\mathcal{F} = \sigma(X)$. Compute $\mathbb{E}[X \mid \mathcal{F}]$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 33 – Conditioning on full information

Solution: Detailed solution

1. Since X is $\sigma(X)$-measurable, X is already known under the information $\mathcal{F}$.
2. Conditional expectation of a known integrable variable is itself.
3. Thus

$$\mathbb{E}[X \mid \sigma(X)] = X.$$

Exercise 34 – Pulling out known factors

Exam-style exercise: Pulling out known factors ⋆⋆

Let Y be bounded and $\mathcal{G}$-measurable, and let X be integrable. State and use the rule for $\mathbb{E}[XY \mid \mathcal{G}]$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 34 – Pulling out known factors

Solution: Detailed solution

1. Since Y is known under $\mathcal{G}$, it can be pulled outside the conditional expectation.
2. The rule is
$$\mathbb{E}[XY \mid \mathcal{G}] = Y\mathbb{E}[X \mid \mathcal{G}].$$
3. Boundedness of Y ensures integrability of XY when X is integrable.
4. This rule is the conditional-expectation analogue of taking a constant outside an ordinary expectation.

Exercise 35 – Tower property

Exam-style exercise: Tower property $\star\star$

Let $\mathcal{G} \subseteq \mathcal{F}$ and let X be integrable. Prove the tower identity $\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{G}]$ in words.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 35 – Tower property

Solution: Detailed solution

1. $\mathbb{E}[X \mid \mathcal{F}]$ is the best summary of X using the larger information $\mathcal{F}$.
2. Conditioning again on the smaller information $\mathcal{G}$ removes the details not visible in $\mathcal{G}$.
3. The resulting variable is $\mathcal{G}$-measurable.
4. It has the correct integrals over all events in $\mathcal{G}$, because $\mathcal{G} \subseteq \mathcal{F}$.
5. By uniqueness of conditional expectation, it equals $\mathbb{E}[X \mid \mathcal{G}]$.

Exercise 36 – Conditional expectation with independent variable

Exam-style exercise: Conditional expectation with independent variable ⋆⋆

Let X be integrable and independent of $\mathcal{G}$. Compute $\mathbb{E}[X \mid \mathcal{G}]$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 36 – Conditional expectation with independent variable

Solution: Detailed solution

1. If X is independent of $\mathcal{G}$, then the information $\mathcal{G}$ does not change the prediction of X.
2. Therefore the conditional expectation is constant.
3. The constant must have expectation equal to $\mathbb{E}[X]$.
4. Hence

$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X].$$

Exercise 37 – Martingale definition check

Exam-style exercise: Martingale definition check ⋆⋆

State the three conditions needed to verify that (M_n) is a martingale with respect to $(\mathcal{F}_n)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 37 – Martingale definition check

Solution: Detailed solution

1. Adaptation: M_n must be $\mathcal{F}_n$-measurable for each n.
2. Integrability: $\mathbb{E}[|M_n|] < \infty$ for each n.
3. Martingale identity:
$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n.$$
4. Missing any one of these conditions is enough for the martingale proof to fail.

Exercise 38 – Symmetric random walk martingale

Exam-style exercise: Symmetric random walk martingale ⋆⋆

Let $S_n = \sum_{k=1}^{n} \xi_k$, where ξ_k are independent, $\mathbb{P}(\xi_k = 1) = \mathbb{P}(\xi_k = -1) = 1/2$, and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$. Show that (S_n) is a martingale.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 38 – Symmetric random walk martingale

Solution: Detailed solution

1. S_n is adapted because it is a function of $\xi_1, \dots, \xi_n$.
2. S_n is integrable because it is bounded by n in absolute value.
3. Compute
$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[S_n + \xi_{n+1} \mid \mathcal{F}_n].$$
4. Since S_n is $\mathcal{F}_n$-measurable, it comes out. Since ξ_{n+1} is independent of $\mathcal{F}_n$ and has mean zero,
$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + 0 = S_n.$$
5. Hence (S_n) is a martingale.

Exercise 39 – Biased random walk classification

Exam-style exercise: Biased random walk classification ⋆⋆

Let S_n be a random walk with $\mathbb{P}(\xi_k = 1) = p$. Classify (S_n) as a martingale, submartingale or supermartingale according to p.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 39 – Biased random walk classification

Solution: Detailed solution

1. We have
$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + \mathbb{E}[\xi_{n+1}] = S_n + (2p - 1).$$
2. If $p = 1/2$, the conditional expectation equals S_n: martingale.
3. If $p > 1/2$, the conditional expectation is at least S_n: submartingale.
4. If $p < 1/2$, it is at most S_n: supermartingale.

Exercise 40 – Centered random walk

Exam-style exercise: Centered random walk ⋆⋆

For a biased random walk, define $M_n = S_n - n(2p-1)$. Show that (M_n) is a martingale.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 40 – Centered random walk

Solution: Detailed solution

1. Adaptation and integrability are immediate from S_n.
2. Compute
$$M_{n+1} = S_n + \xi_{n+1} - (n+1)(2p-1).$$
3. Conditional on $\mathcal{F}_n$,
$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = S_n + (2p-1) - (n+1)(2p-1).$$
4. Simplify:
$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = S_n - n(2p-1) = M_n.$$
5. Thus centering removes the drift.

Exercise 41 – Square minus variance martingale

Exam-style exercise: Square minus variance martingale $\star\star\star$

For the symmetric random walk, show that $M_n = S_n^2 - n$ is a martingale.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 41 – Square minus variance martingale

Solution: Detailed solution

1. Write $S_{n+1} = S_n + \xi_{n+1}$.
2. Square:
$$S_{n+1}^2 = S_n^2 + 2S_n\xi_{n+1} + \xi_{n+1}^2.$$
3. Conditional on $\mathcal{F}_n$, S_n is known, $\mathbb{E}[\xi_{n+1}] = 0$, and $\xi_{n+1}^2 = 1$.
4. Hence
$$\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = S_n^2 + 1.$$
5. Therefore
$$\mathbb{E}[S_{n+1}^2 - (n+1) \mid \mathcal{F}_n] = S_n^2 + 1 - n - 1 = S_n^2 - n.$$
6. Thus M_n is a martingale.

Exercise 42 – Exponential martingale in discrete time

Exam-style exercise: Exponential martingale in discrete time $\star\star\star$

Let (ξ_k) be i.i.d. and let $m(\theta) = \mathbb{E}[e^{\theta \xi_1}] < \infty$. Show that $M_n = e^{\theta S_n}/m(\theta)^n$ is a martingale.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 42 – Exponential martingale in discrete time

Solution: Detailed solution

1. The process is adapted and nonnegative.
2. Since $S_{n+1} = S_n + \xi_{n+1}$,
$$M_{n+1} = \frac{e^{\theta S_n} e^{\theta \xi_{n+1}}}{m(\theta)^{n+1}}.$$
3. Conditional on $\mathcal{F}_n$, $e^{\theta S_n}$ is known and ξ_{n+1} is independent of $\mathcal{F}_n$.
4. Therefore
$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = \frac{e^{\theta S_n}}{m(\theta)^{n+1}} \mathbb{E}[e^{\theta \xi_{n+1}}] = \frac{e^{\theta S_n}}{m(\theta)^n} = M_n.$$

Exercise 43 – Convex transformation

Exam-style exercise: Convex transformation ⋆⋆⋆

Let (M_n) be a martingale and let φ be convex with suitable integrability. Show that $(\varphi(M_n))$ is a submartingale.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 43 – Convex transformation

Solution: Detailed solution

1. Since M is a martingale,
$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n.$$
2. By conditional Jensen's inequality,
$$\mathbb{E}[\varphi(M_{n+1}) \mid \mathcal{F}_n] \geq \varphi(\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]).$$
3. Substitute the martingale identity:
$$\mathbb{E}[\varphi(M_{n+1}) \mid \mathcal{F}_n] \geq \varphi(M_n).$$
4. Hence $\varphi(M_n)$ is a submartingale.

Module 4 Exercise Pack

Martingales, transforms and examples

Exercise 44 – Martingale transform

Exam-style exercise: Martingale transform $\star\star\star$

Let (M_n) be a martingale and let (H_n) be bounded and predictable. Define $N_n = \sum_{k=1}^{n} H_k(M_k - M_{k-1})$. Show that (N_n) is a martingale.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 44 – Martingale transform

Solution: Detailed solution

1. N_n is adapted because each H_k is known at time $k-1$ and $M_k - M_{k-1}$ is known at time k.
2. Boundedness of H_k and integrability of M_k give integrability.
3. Compute one-step conditional expectation:

$$N_{n+1} = N_n + H_{n+1}(M_{n+1} - M_n).$$

4. Since H_{n+1} is $\mathcal{F}_n$-measurable,

$$\mathbb{E}[N_{n+1} \mid \mathcal{F}_n] = N_n + H_{n+1}\mathbb{E}[M_{n+1} - M_n \mid \mathcal{F}_n] = N_n.$$

5. Hence N is a martingale.

Exercise 45 – A nonnegative martingale

Exam-style exercise: A nonnegative martingale ⋆⋆

Let $M_n = \prod_{k=1}^{n} Z_k$, where $Z_k > 0$ are i.i.d., independent of the past, and $\mathbb{E}[Z_k] = 1$. Show that (M_n) is a martingale.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 45 – A nonnegative martingale

Solution: Detailed solution

1. M_n is adapted to $\mathcal{F}_n = \sigma(Z_1, \dots, Z_n)$.
2. It is integrable because $\mathbb{E}[M_n] = \prod_{k=1}^{n} \mathbb{E}[Z_k] = 1$.
3. Since $M_{n+1} = M_n Z_{n+1}$,
$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \mathbb{E}[Z_{n+1} \mid \mathcal{F}_n].$$
4. By independence, $\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[Z_{n+1}] = 1$.
5. Therefore $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$.

Exercise 46 – Doob decomposition in one step

Exam-style exercise: Doob decomposition in one step ⋆⋆⋆

Let (X_n) be an integrable submartingale. Define $A_0 = 0$ and $A_n = \sum_{k=1}^{n} \mathbb{E}[X_k - X_{k-1} \mid \mathcal{F}_{k-1}]$. Explain why A_n is increasing and why $M_n = X_n - A_n$ is a martingale.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 46 – Doob decomposition in one step

Solution: Detailed solution

1. Submartingale means
$$\mathbb{E}[X_k \mid \mathcal{F}_{k-1}] \geq X_{k-1}.$$
2. Hence
$$\mathbb{E}[X_k - X_{k-1} \mid \mathcal{F}_{k-1}] \geq 0.$$
3. Therefore the increments of A are nonnegative, so A is increasing.
4. Now compute
$$M_k - M_{k-1} = X_k - X_{k-1} - (A_k - A_{k-1}).$$
5. Conditional expectation given $\mathcal{F}_{k-1}$ is zero by the definition of $A_k - A_{k-1}$.
6. Thus M is a martingale.

Exercise 47 – Submartingale from square

Exam-style exercise: Submartingale from square $\star\star$

Let (M_n) be a square-integrable martingale. Show that (M_n^2) is a submartingale.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 47 – Submartingale from square

Solution: Detailed solution

1. The function $\varphi(x) = x^2$ is convex.
2. By conditional Jensen,
$$\mathbb{E}[M_{n+1}^2 \mid \mathcal{F}_n] \geq (\mathbb{E}[M_{n+1} \mid \mathcal{F}_n])^2.$$
3. Since M is a martingale,
$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n.$$
4. Therefore
$$\mathbb{E}[M_{n+1}^2 \mid \mathcal{F}_n] \geq M_n^2.$$
5. So M_n^2 is a submartingale.

Exercise 48 – Supermartingale from negative drift

Exam-style exercise: Supermartingale from negative drift $\star\star$

Let $X_{n+1} = X_n + \eta_{n+1}$ with $\mathbb{E}[\eta_{n+1} \mid \mathcal{F}_n] \leq 0$. Show that (X_n) is a supermartingale, assuming integrability and adaptation.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 48 – Supermartingale from negative drift

Solution: Detailed solution

1. Compute
$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[X_n + \eta_{n+1} \mid \mathcal{F}_n].$$
2. Since X_n is $\mathcal{F}_n$-measurable,
$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n + \mathbb{E}[\eta_{n+1} \mid \mathcal{F}_n].$$
3. The conditional drift is nonpositive.
4. Hence
$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n.$$
5. Therefore X is a supermartingale.

Exercise 49 – Testing a proposed martingale

Exam-style exercise: Testing a proposed martingale ⋆⋆⋆

Let $M_n = S_n^3 - 3nS_n$ for a symmetric random walk. Is (M_n) a martingale?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 49 – Testing a proposed martingale

Solution: Detailed solution

1. Write $S_{n+1} = S_n + \xi$, where $\xi = \xi_{n+1}$ is independent of $\mathcal{F}_n$, $\mathbb{E}[\xi] = 0$, $\mathbb{E}[\xi^2] = 1$, $\mathbb{E}[\xi^3] = 0$.
2. Expand:
$$(S_n + \xi)^3 = S_n^3 + 3S_n^2\xi + 3S_n\xi^2 + \xi^3.$$
3. Conditional expectation gives
$$\mathbb{E}[S_{n+1}^3 \mid \mathcal{F}_n] = S_n^3 + 3S_n.$$
4. Therefore
$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = S_n^3 + 3S_n - 3(n+1)S_n = S_n^3 - 3nS_n = M_n.$$
5. Thus it is a martingale.

Exercise 50 – Non-integrable trap

Exam-style exercise: Non-integrable trap ⋆⋆

A process satisfies $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$ formally, but $\mathbb{E}[|M_n|] = \infty$ for some n. Is it a martingale?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 50 – Non-integrable trap

Solution: Detailed solution

1. No. Integrability is part of the definition of a martingale.
2. Conditional expectations are defined for integrable random variables in the standard setting.
3. Therefore the equality alone is not enough.
4. The correct conclusion is: the process is not a martingale unless all M_n are integrable.

Exercise 51 – Constant expectation of martingale

Exam-style exercise: Constant expectation of martingale $\star\star$

Let (M_n) be a martingale. Show that $\mathbb{E}[M_n] = \mathbb{E}[M_0]$ for all n.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 51 – Constant expectation of martingale

Solution: Detailed solution

1. Use the tower property:
$$\mathbb{E}[M_{n+1}] = \mathbb{E}[\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]].$$
2. By the martingale property, $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$.
3. Hence $\mathbb{E}[M_{n+1}] = \mathbb{E}[M_n]$.
4. Iterating gives $\mathbb{E}[M_n] = \mathbb{E}[M_0]$ for all n.

Exercise 52 – A process with constant mean but not martingale

Exam-style exercise: A process with constant mean but not martingale ⋆⋆⋆

Let $X_0 = 0$, $X_1 = \xi$, $X_2 = -\xi$, where ξ takes values ± 1 with probability $1/2$. With the natural filtration, does constant mean imply martingale?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 52 – A process with constant mean but not martingale

Solution: Detailed solution

1. The means are $\mathbb{E}[X_0] = 0$, $\mathbb{E}[X_1] = 0$, $\mathbb{E}[X_2] = 0$.
2. So the expectation is constant.
3. However,
$$\mathbb{E}[X_2 \mid \mathcal{F}_1] = \mathbb{E}[-\xi \mid \sigma(\xi)] = -\xi.$$
4. But $X_1 = \xi$.
5. Since $-\xi \neq \xi$ except on no event of positive probability, the martingale identity fails.
6. Constant mean is necessary but not sufficient for being a martingale.

Exercise 53 – Submartingale expectation

Exam-style exercise: Submartingale expectation $\star\star$

If (X_n) is a submartingale, show that $\mathbb{E}[X_n]$ is nondecreasing.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 53 – Submartingale expectation

Solution: Detailed solution

1. By definition,
$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n.$$
2. Take expectations on both sides:
$$\mathbb{E}[\mathbb{E}[X_{n+1} \mid \mathcal{F}_n]] \geq \mathbb{E}[X_n].$$
3. Use the tower property:
$$\mathbb{E}[X_{n+1}] \geq \mathbb{E}[X_n].$$
4. Hence the mean is nondecreasing.

Module 4 Exercise Pack

Stopping times and optional sampling

Exercise 54 – First hitting time

Exam-style exercise: First hitting time ⋆⋆

Let $\tau_a = \inf\{n \geq 0 : X_n \geq a\}$ for an adapted discrete-time process. Show that τ_a is a stopping time.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 54 – First hitting time

Solution: Detailed solution

1. We must show $\{\tau_a \leq n\} \in \mathcal{F}_n$ for every n.
2. As before,
$$\{\tau_a \leq n\} = \bigcup_{k=0}^{n} \{X_k \geq a\}.$$
3. Since X_k is $\mathcal{F}_k$-measurable and $\mathcal{F}_k \subseteq \mathcal{F}_n$, each event belongs to $\mathcal{F}_n$.
4. A finite union of events in $\mathcal{F}_n$ is in $\mathcal{F}_n$.
5. Hence τ_a is a stopping time.

Exercise 55 – Last hitting time is not stopping

Exam-style exercise: Last hitting time is not stopping $\star\star\star$

Let $\sigma = \sup\{0 \le k \le N : X_k = 0\}$ for a fixed horizon N. Explain why σ is generally not a stopping time before time N.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 55 – Last hitting time is not stopping

Solution: Detailed solution

1. To know whether $\sigma \leq n$ with $n < N$, one must know that no zero occurs after time n up to time N.
2. This requires information about $X_{n+1}, \ldots, X_N$.
3. That information is not contained in $\mathcal{F}_n$.
4. Therefore $\{\sigma \leq n\}$ is generally not $\mathcal{F}_n$-measurable.
5. Last occurrence times usually depend on the future and are not stopping times.

Exercise 56 – Minimum of stopping times

Exam-style exercise: Minimum of stopping times ⋆⋆

If τ and σ are stopping times, show that $\tau \wedge \sigma$ is a stopping time.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 56 – Minimum of stopping times

Solution: Detailed solution

1. We need to check $\{\tau \wedge \sigma \leq n\} \in \mathcal{F}_n$.
2. The minimum is at most n if at least one of the two times is at most n:

$$\{\tau \wedge \sigma \leq n\} = \{\tau \leq n\} \cup \{\sigma \leq n\}.$$

3. Since τ and σ are stopping times, both events are in $\mathcal{F}_n$.
4. Their union is in $\mathcal{F}_n$.
5. Therefore $\tau \wedge \sigma$ is a stopping time.

Exercise 57 – Maximum of stopping times

Exam-style exercise: Maximum of stopping times ⋆⋆

If τ and σ are stopping times, show that $\tau \vee \sigma$ is a stopping time.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 57 – Maximum of stopping times

Solution: Detailed solution

1. Check
$$\{\tau \vee \sigma \leq n\} = \{\tau \leq n\} \cap \{\sigma \leq n\}.$$
2. Both events are in $\mathcal{F}_n$ by the stopping-time property.
3. Sigma-algebras are closed under finite intersections.
4. Hence $\tau \vee \sigma$ is a stopping time.

Exercise 58 – Bounded optional sampling

Exam-style exercise: Bounded optional sampling $\star\star\star$

Let (M_n) be a martingale and let τ be a bounded stopping time with $\tau \leq N$. State the optional sampling conclusion and interpret it.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 58 – Bounded optional sampling

Solution: Detailed solution

1. Under boundedness, optional sampling gives

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

2. The stopped value has the same expectation as the initial value.
3. The condition $\tau \leq N$ prevents integrability problems due to unbounded waiting.
4. Intuition: a fair martingale remains fair when stopped by a rule that uses only current and past information and is bounded.

Exercise 59 – Stopped martingale

Exam-style exercise: Stopped martingale $\star\star\star$

Let M be a martingale and τ a stopping time. Show that $M_{n\wedge\tau}$ is a martingale in discrete time, assuming integrability.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 59 – Stopped martingale

Solution: Detailed solution

1. Write the increment:

$$M_{(n+1)\wedge\tau} - M_{n\wedge\tau} = \mathbf{1}_{\{\tau>n\}}(M_{n+1} - M_n).$$

2. The event $\{\tau > n\}$ is $\mathcal{F}_n$-measurable because τ is a stopping time.
3. Hence its indicator is predictable.
4. Conditional expectation:

$$\mathbb{E}[\mathbf{1}_{\{\tau>n\}}(M_{n+1} - M_n) \mid \mathcal{F}_n] = \mathbf{1}_{\{\tau>n\}} \cdot 0 = 0.$$

5. Therefore the stopped process is a martingale.

Exercise 60 – Gambler's ruin probability

Exam-style exercise: Gambler's ruin probability $\star\star\star\star$

Let (S_n) be the symmetric random walk starting from $i \in \{0, \dots, N\}$. Let $\tau = \inf\{n : S_n = 0 \text{ or } S_n = N\}$. Compute $\mathbb{P}_i(S_\tau = N)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 60 – Gambler's ruin probability

Solution: Detailed solution

1. The symmetric random walk is a martingale.
2. The stopping time τ is bounded in expectation and standard optional stopping applies after localization; in the finite ruin problem this is valid.
3. Since S_τ is either 0 or N,
$$\mathbb{E}_i[S_\tau] = N\mathbb{P}_i(S_\tau = N).$$
4. Optional stopping gives $\mathbb{E}_i[S_\tau] = \mathbb{E}_i[S_0] = i$.
5. Therefore
$$\mathbb{P}_i(S_\tau = N) = \frac{i}{N}.$$

Exercise 61 – Expected ruin time

Exam-style exercise: Expected ruin time ⋆⋆⋆⋆⋆

For the same symmetric random walk on $\{0, \dots, N\}$, use $S_n^2 - n$ to compute $\mathbb{E}_i[\tau]$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 61 – Expected ruin time

Solution: Detailed solution

1. The process $S_n^2 - n$ is a martingale.
2. Optional stopping gives
$$\mathbb{E}_i[S_\tau^2 - \tau] = S_0^2 = i^2.$$
3. Since $S_\tau \in \{0, N\}$,
$$\mathbb{E}_i[S_\tau^2] = N^2 \mathbb{P}_i(S_\tau = N).$$
4. From the previous exercise, $\mathbb{P}_i(S_\tau = N) = i/N$, so
$$\mathbb{E}_i[S_\tau^2] = N^2 \frac{i}{N} = Ni.$$
5. Therefore
$$Ni - \mathbb{E}_i[\tau] = i^2, \qquad \mathbb{E}_i[\tau] = i(N - i).$$

Exercise 62 – Why optional stopping can fail

Exam-style exercise: Why optional stopping can fail $\star\star\star$

Let S_n be a symmetric random walk and $\tau = \inf\{n : S_n = 1\}$. Why can one not conclude $\mathbb{E}[S_\tau] = \mathbb{E}[S_0] = 0$?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 62 – Why optional stopping can fail

Solution: Detailed solution

1. Here $S_\tau = 1$ on the event $\tau < \infty$, which has probability one for the symmetric random walk.
2. Thus $\mathbb{E}[S_\tau] = 1$, not 0.
3. The stopping time is unbounded.
4. The hypotheses of the bounded optional sampling theorem are not satisfied.
5. This example shows that optional stopping requires conditions, not only the martingale property.

Exercise 63 – Stopping time with deterministic horizon

Exam-style exercise: Stopping time with deterministic horizon $\star\star$

If τ is a stopping time and N is deterministic, show that $\tau \wedge N$ is a bounded stopping time.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 63 – Stopping time with deterministic horizon

Solution: Detailed solution

1. Since $\tau \wedge N \leq N$, it is bounded by a deterministic constant.
2. It is the minimum of the stopping time τ and the deterministic stopping time N.
3. The minimum of stopping times is a stopping time.
4. Therefore $\tau \wedge N$ is bounded and is a stopping time.

Module 4 Exercise Pack

Discrete-time Markov chains

Exercise 64 – Markov property

Exam-style exercise: Markov property ⋆⋆

State the Markov property for a discrete-time process with countable state space.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 64 – Markov property

Solution: Detailed solution

1. The process is Markov if, for every state j,

$$\mathbb{P}(X_{n+1} = j \mid X_0, \ldots, X_n) = \mathbb{P}(X_{n+1} = j \mid X_n).$$

2. The future depends on the past only through the present state.
3. For a time-homogeneous chain, the right-hand side depends only on X_n and not on n.

Exercise 65 – Transition matrix calculation

Exam-style exercise: Transition matrix calculation ⋆⋆

Let

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix}.$$

If the initial distribution is $\mu_0 = (0.4, 0.6)$, compute μ_1.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 65 – Transition matrix calculation

Solution: Detailed solution

1. For a row-vector convention, the next distribution is $\mu_1 = \mu_0 P$.
2. Compute the first coordinate:

$$0.4 \cdot 0.7 + 0.6 \cdot 0.2 = 0.28 + 0.12 = 0.40.$$

3. Compute the second coordinate:

$$0.4 \cdot 0.3 + 0.6 \cdot 0.8 = 0.12 + 0.48 = 0.60.$$

4. Therefore $\mu_1 = (0.4, 0.6)$.
5. In this case the initial distribution is already invariant.

Exercise 66 – Two-step transition

Exam-style exercise: Two-step transition ⋆⋆

For

$$P = \begin{pmatrix} 1/2 & 1/2 \\ 1/4 & 3/4 \end{pmatrix},$$

compute the probability of going from state 1 to state 2 in two steps.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 66 – Two-step transition

Solution: Detailed solution

1. The two-step transition probabilities are entries of P^2.
2. The desired entry is $(P^2)_{12}$.
3. Compute row 1 times column 2:

$$(P^2)_{12} = \frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{3}{4} = \frac{1}{4} + \frac{3}{8} = \frac{5}{8}.$$

4. Therefore $\mathbb{P}(X_2 = 2 \mid X_0 = 1) = 5/8$.

Exercise 67 – Invariant distribution

Exam-style exercise: Invariant distribution $\star\star\star$

Find the invariant distribution of

$$P = \begin{pmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{pmatrix}.$$

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 67 – Invariant distribution

Solution: Detailed solution

1. Let $\pi = (\pi_1, \pi_2)$ with $\pi_1 + \pi_2 = 1$.
2. Solve $\pi = \pi P$.
3. The first coordinate gives
$$\pi_1 = 0.9\pi_1 + 0.4\pi_2.$$
4. Hence $0.1\pi_1 = 0.4\pi_2$, so $\pi_1 = 4\pi_2$.
5. With $\pi_1 + \pi_2 = 1$, we get $5\pi_2 = 1$, so $\pi_2 = 1/5$, $\pi_1 = 4/5$.
6. Therefore $\pi = (4/5, 1/5)$.

Exercise 68 – Absorbing state

Exam-style exercise: Absorbing state ⋆⋆

Consider

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0.3 & 0.4 & 0.3 \\ 0 & 0 & 1 \end{pmatrix}.$$

Which states are absorbing?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 68 – Absorbing state

Solution: Detailed solution

1. A state i is absorbing if $P_{ii} = 1$.
2. State 1 has row $(1, 0, 0)$, so once entered it cannot be left.
3. State 3 has row $(0, 0, 1)$, so it is also absorbing.
4. State 2 is not absorbing because its row is not concentrated on state 2.
5. Therefore states 1 and 3 are absorbing.

Exercise 69 – Hitting probability in a chain

Exam-style exercise: Hitting probability in a chain $\star\star\star$

For the Markov chain on $\{0, 1, 2\}$ with absorbing states 0 and 2, and transition probabilities $p_{10} = 0.3$, $p_{11} = 0.2$, $p_{12} = 0.5$, compute the probability of hitting 2 before 0 starting from 1.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 69 – Hitting probability in a chain

Solution: Detailed solution

1. Let $h_i = \mathbb{P}_i(T_2 < T_0)$.
2. Boundary conditions: $h_0 = 0$, $h_2 = 1$.
3. Starting from 1, use first-step analysis:

$$h_1 = 0.3h_0 + 0.2h_1 + 0.5h_2.$$

4. Substitute the boundary values:

$$h_1 = 0.2h_1 + 0.5.$$

5. Thus $0.8h_1 = 0.5$, so

$$h_1 = \frac{5}{8}.$$

Exercise 70 – Expected hitting time

Exam-style exercise: Expected hitting time ⋆⋆⋆⋆

In the previous chain, compute the expected time to hit either 0 or 2 starting from 1.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 70 – Expected hitting time

Solution: Detailed solution

1. Let $e_i = \mathbb{E}_i[T_{\{0,2\}}]$.
2. Boundary conditions: $e_0 = e_2 = 0$.
3. From state 1, one step is spent, then the chain moves according to the transition probabilities:
$$e_1 = 1 + 0.3e_0 + 0.2e_1 + 0.5e_2.$$
4. Substitute $e_0 = e_2 = 0$:
$$e_1 = 1 + 0.2e_1.$$
5. Hence $0.8e_1 = 1$, so
$$e_1 = \frac{5}{4}.$$

Exercise 71 – Chapman–Kolmogorov

Exam-style exercise: Chapman–Kolmogorov ⋆⋆⋆

Explain why $p_{ij}^{(m+n)} = \sum_k p_{ik}^{(m)} p_{kj}^{(n)}$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 71 – Chapman–Kolmogorov

Solution: Detailed solution

1. To go from i to j in $m+n$ steps, condition on the state k occupied after m steps.
2. The events $\{X_m = k\}$ form a partition of the state space.
3. Thus
$$\mathbb{P}_i(X_{m+n} = j) = \sum_k \mathbb{P}_i(X_m = k, X_{m+n} = j).$$
4. By the Markov property, after reaching k at time m, the probability of being at j after another n steps is $p_{kj}^{(n)}$.
5. Therefore
$$p_{ij}^{(m+n)} = \sum_k p_{ik}^{(m)} p_{kj}^{(n)}.$$

Exercise 72 – Irreducibility

Exam-style exercise: Irreducibility ⋆⋆

For

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.2 & 0.5 & 0.3 \\ 0 & 0.4 & 0.6 \end{pmatrix},$$

is the chain irreducible?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 72 – Irreducibility

Solution: Detailed solution

1. A finite chain is irreducible if every state can be reached from every other state, possibly in several steps.
2. From state 1, one can go to 2; from 2, one can go to 3; hence 1 can reach 3.
3. From state 3, one can go to 2; from 2, one can go to 1; hence 3 can reach 1.
4. State 2 communicates directly with both 1 and 3.
5. Therefore all states communicate: the chain is irreducible.

Exercise 73 – Period

Exam-style exercise: Period ⋆⋆⋆

Consider a two-state chain with transition matrix

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Find the period of each state.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 73 – Period

Solution: Detailed solution

1. Starting from state 1, the chain returns to 1 after 2, 4, 6, ... steps.
2. The set of possible return times is $\{2, 4, 6, \ldots\}$.
3. The greatest common divisor is 2.
4. Thus state 1 has period 2.
5. By symmetry, state 2 also has period 2.

Exercise 74 – Aperiodicity via self-loop

Exam-style exercise: Aperiodicity via self-loop ⋆⋆

Show that if an irreducible finite Markov chain has one state i with $p_{ii} > 0$, then all states have period one.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 74 – Aperiodicity via self-loop

Solution: Detailed solution

1. If $p_{ii} > 0$, state i can return to itself in one step.
2. Therefore the period of i divides 1, so it is 1.
3. In an irreducible chain, all states have the same period.
4. Hence every state has period 1.
5. The chain is aperiodic.

Exercise 75 – Classify recurrent in finite irreducible chain

Exam-style exercise: Classify recurrent in finite irreducible chain ⋆⋆

What can be said about recurrence of states in a finite irreducible Markov chain?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 75 – Classify recurrent in finite irreducible chain

Solution: Detailed solution

1. In a finite irreducible chain, there is only one communicating class.
2. The chain cannot escape to infinitely many new states because the state space is finite.
3. Standard Markov-chain theory implies every state is positive recurrent.
4. Therefore each state is visited infinitely often with probability one and has finite mean return time.

Exercise 76 – Long-run distribution

Exam-style exercise: Long-run distribution $\star\star\star$

Assume a finite Markov chain is irreducible and aperiodic with invariant distribution π. What is the limiting distribution of X_n?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 76 – Long-run distribution

Solution: Detailed solution

1. For a finite irreducible aperiodic chain, the convergence theorem applies.
2. Regardless of the initial state or initial distribution,

$$\mathbb{P}(X_n = j) \to \pi_j.$$

3. Equivalently, rows of P^n converge to π.
4. The invariant distribution describes the long-run fraction of time spent in each state.

Exercise 77 – Markov property false example

Exam-style exercise: Markov property false example ⋆⋆⋆

Let $X_n = \xi_n + \xi_{n-1}$, where (ξ_n) are i.i.d. Bernoulli variables. Explain why (X_n) is generally not Markov.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 77 – Markov property false example

Solution: Detailed solution

1. The present value X_n is the sum of two consecutive innovations.
2. Knowing only X_n may not reveal the individual value of ξ_n.
3. But $X_{n+1} = \xi_{n+1} + \xi_n$ depends on ξ_n.
4. Additional past information can improve knowledge of ξ_n and therefore affect the distribution of X_{n+1}.
5. Hence the conditional law of the future is not determined by X_n alone in general.

Module 4 Exercise Pack

Continuous-time Markov dynamics

Exercise 78 – Semigroup property

Exam-style exercise: Semigroup property $\star\star\star$

Let $(P_t)_{t\geq 0}$ be the transition semigroup of a time-homogeneous Markov process. State and interpret $P_{t+s} = P_t P_s$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 78 – Semigroup property

Solution: Detailed solution

1. The semigroup identity is
$$P_{t+s} = P_t P_s.$$
2. Moving forward by $t+s$ units of time is equivalent to first moving by s and then by t.
3. This follows from the Markov property: once the intermediate state is known, the future depends only on that state.
4. It is the continuous-time analogue of $P^{m+n} = P^m P^n$ for discrete Markov chains.

Exercise 79 – Generator from semigroup

Exam-style exercise: Generator from semigroup ⋆⋆⋆

Define the generator A of a Markov semigroup (P_t) and explain its meaning.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 79 – Generator from semigroup

Solution: Detailed solution

1. The generator is formally
$$Af = \lim_{t \downarrow 0} \frac{P_t f - f}{t},$$
for functions f for which the limit exists.
2. It describes the instantaneous rate of change of the expected value of $f(X_t)$.
3. If $P_t f(x) = \mathbb{E}_x[f(X_t)]$, then
$$Af(x) = \frac{d}{dt} P_t f(x) \Big|_{t=0}.$$
4. Thus the generator is the local description of the Markov dynamics.

Exercise 80 – Generator of two-state chain

Exam-style exercise: Generator of two-state chain ⋆⋆⋆

Let a continuous-time chain on $\{0,1\}$ jump from 0 to 1 at rate λ and from 1 to 0 at rate μ. Write its generator matrix.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 80 – Generator of two-state chain

Solution: Detailed solution

1. Off-diagonal entries are transition rates.
2. Thus $q_{01} = \lambda$ and $q_{10} = \mu$.
3. Rows of a generator sum to zero.
4. Therefore $q_{00} = -\lambda$ and $q_{11} = -\mu$.
5. The generator is

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}.$$

Exercise 81 – Kolmogorov forward equation

Exam-style exercise: Kolmogorov forward equation $\star\star\star$

For a finite-state continuous-time Markov chain with generator Q, state the forward equation for the row-vector distribution μ_t.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 81 – Kolmogorov forward equation

Solution: Detailed solution

1. If distributions are written as row vectors, the forward equation is
$$\frac{d}{dt}\mu_t = \mu_t Q.$$
2. It describes how probabilities of states change over time.
3. The initial condition is μ_0.
4. The formal solution is
$$\mu_t = \mu_0 e^{tQ}.$$

Exercise 82 – Kolmogorov backward equation

Exam-style exercise: Kolmogorov backward equation $\star\star\star$

For $u(t,x) = \mathbb{E}_x[f(X_t)]$, state the backward equation in terms of the generator A.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 82 – Kolmogorov backward equation

Solution: Detailed solution

1. Since $u(t,x) = P_t f(x)$, the semigroup evolves according to its generator.
2. The backward equation is
$$\partial_t u(t,x) = Au(t,\cdot)(x).$$
3. The initial condition is
$$u(0,x) = f(x).$$
4. It is called backward because the generator acts on the starting state variable.

Exercise 83 – Feller intuition

Exam-style exercise: Feller intuition ⋆⋆

In simple words, what does the Feller property preserve?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 83 – Feller intuition

Solution: Detailed solution

1. The Feller property says that the semigroup maps nice continuous functions into nice continuous functions.
2. Typically, if f is continuous and vanishes at infinity, then $P_t f$ has the same property.
3. Intuitively: small changes in the starting point produce small changes in expected future values.
4. It is a regularity property of the Markov dynamics.

Exercise 84 – Deterministic flow as Markov process

Exam-style exercise: Deterministic flow as Markov process $\star\star\star$

Let $X_t = x + t$ when $X_0 = x$. Is this deterministic process Markov? Find $P_t f(x)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 84 – Deterministic flow as Markov process

Solution: Detailed solution

1. Even deterministic dynamics can be Markov because the future is determined by the present.
2. Starting from x, after time t the state is $x + t$.
3. Therefore
$$P_t f(x) = f(x + t).$$
4. The future does not require knowledge of the earlier path, only the present state.
5. Hence the process is Markov.

Exercise 85 – Generator of deterministic flow

Exam-style exercise: Generator of deterministic flow $\star\star\star$

For the deterministic flow $X_t = x + t$, compute the generator on differentiable functions.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 85 – Generator of deterministic flow

Solution: Detailed solution

1. We found $P_t f(x) = f(x+t)$.
2. Therefore
$$Af(x) = \lim_{t \downarrow 0} \frac{f(x+t) - f(x)}{t}.$$
3. If f is differentiable, this limit is $f'(x)$.
4. Hence the generator is
$$Af = f'.$$

Module 4 Exercise Pack

Brownian motion

Exercise 86 – Brownian increments

Exam-style exercise: Brownian increments ⋆⋆

Let (B_t) be Brownian motion. Find the distribution of $B_t - B_s$ for $0 \leq s < t$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 86 – Brownian increments

Solution: Detailed solution

1. Brownian motion has stationary independent Gaussian increments.
2. The increment over an interval of length $t - s$ is normal with mean zero.
3. Its variance equals the length of the interval.
4. Therefore

$$B_t - B_s \sim N(0, t - s).$$

Exercise 87 – Covariance of Brownian motion

Exam-style exercise: Covariance of Brownian motion ⋆⋆⋆

Show that $\mathrm{Cov}(B_s, B_t) = \min\{s, t\}$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 87 – Covariance of Brownian motion

Solution: Detailed solution

1. Assume $s \le t$.
2. Write
$$B_t = B_s + (B_t - B_s).$$
3. The increment $B_t - B_s$ is independent of B_s and has mean zero.
4. Thus
$$\mathrm{Cov}(B_s, B_t) = \mathrm{Cov}(B_s, B_s) + \mathrm{Cov}(B_s, B_t - B_s) = \mathrm{Var}(B_s) + 0 = s.$$
5. By symmetry, the answer is $\min\{s, t\}$.

Exercise 88 – Brownian martingale

Exam-style exercise: Brownian martingale ⋆⋆

Show that Brownian motion is a martingale with respect to its natural filtration.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 88 – Brownian martingale

Solution: Detailed solution

1. Brownian motion is adapted to its natural filtration.
2. It is integrable because $B_t \sim N(0, t)$.
3. For $s < t$,
$$B_t = B_s + (B_t - B_s).$$
4. The increment is independent of $\mathcal{F}_s$ and has mean zero.
5. Hence
$$\mathbb{E}[B_t \mid \mathcal{F}_s] = B_s.$$
6. Therefore Brownian motion is a martingale.

Exercise 89 – Brownian square martingale

Exam-style exercise: Brownian square martingale ⋆⋆⋆

Show that $B_t^2 - t$ is a martingale.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 89 – Brownian square martingale

Solution: Detailed solution

1. For $s < t$, write $B_t = B_s + \Delta$, where $\Delta = B_t - B_s \sim N(0, t-s)$ and is independent of $\mathcal{F}_s$.
2. Expand:
$$B_t^2 = B_s^2 + 2B_s\Delta + \Delta^2.$$
3. Conditional on $\mathcal{F}_s$, B_s is known, $\mathbb{E}[\Delta] = 0$, and $\mathbb{E}[\Delta^2] = t - s$.
4. Thus
$$\mathbb{E}[B_t^2 \mid \mathcal{F}_s] = B_s^2 + t - s.$$
5. Therefore
$$\mathbb{E}[B_t^2 - t \mid \mathcal{F}_s] = B_s^2 - s.$$

Exercise 90 – Distribution of scaled Brownian motion

Exam-style exercise: Distribution of scaled Brownian motion ⋆⋆

For fixed $c > 0$, show that $B_{ct}/\sqrt{c}$ has the same one-time distribution as B_t.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 90 – Distribution of scaled Brownian motion

Solution: Detailed solution

1. Since $B_{ct} \sim N(0, ct)$,
$$\frac{B_{ct}}{\sqrt{c}} \sim N\left(0, \frac{ct}{c}\right) = N(0, t).$$
2. But $B_t \sim N(0, t)$.
3. Hence the one-time distributions are the same.
4. In fact, Brownian scaling holds at the process level, not only at one time.

Exercise 91 – Brownian independent increments

Exam-style exercise: Brownian independent increments ⋆⋆

For $0 < t_1 < t_2 < t_3$, are $B_{t_2} - B_{t_1}$ and $B_{t_3} - B_{t_2}$ independent?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 91 – Brownian independent increments

Solution: Detailed solution

1. The intervals $(t_1, t_2]$ and $(t_2, t_3]$ are disjoint.
2. Brownian motion has independent increments over disjoint intervals.
3. Therefore the two increments are independent.
4. Their distributions are $N(0, t_2 - t_1)$ and $N(0, t_3 - t_2)$ respectively.

Exercise 92 – Brownian is not stationary

Exam-style exercise: Brownian is not stationary ⋆⋆

Is Brownian motion strictly stationary? Give a simple reason.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 92 – Brownian is not stationary

Solution: Detailed solution

1. Brownian motion satisfies $B_t \sim N(0, t)$.
2. The variance of B_t is t.
3. Since the distribution changes with time, the one-time marginal laws are not invariant under shifts.
4. Therefore Brownian motion is not strictly stationary.

Exercise 93 – Brownian has stationary increments

Exam-style exercise: Brownian has stationary increments ⋆⋆

Show that Brownian increments are stationary.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 93 – Brownian has stationary increments

Solution: Detailed solution

1. For $h > 0$,
$$B_{t+h} - B_t \sim N(0, h).$$
2. The distribution depends on the interval length h.
3. It does not depend on the starting time t.
4. Therefore Brownian motion has stationary increments.

Exercise 94 – Quadratic variation preview

Exam-style exercise: Quadratic variation preview ⋆⋆⋆

For a partition $0 = t_0 < \cdots < t_n = T$, explain why $\sum_i (B_{t_i} - B_{t_{i-1}})^2$ should be close to T when the mesh is small.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 94 – Quadratic variation preview

Solution: Detailed solution

1. Each Brownian increment over $[t_{i-1}, t_i]$ has variance Δt_i.
2. Hence
$$\mathbb{E}[(B_{t_i} - B_{t_{i-1}})^2] = \Delta t_i.$$
3. Summing expectations gives
$$\mathbb{E}\left[\sum_i (\Delta B_i)^2\right] = \sum_i \Delta t_i = T.$$
4. The quadratic variation theorem states that the random sum converges to T as the mesh goes to zero.
5. This is why Brownian motion has quadratic variation $[B]_T = T$.

Exercise 95 – Brownian hitting probability via martingale

Exam-style exercise: Brownian hitting probability via martingale $\star\star\star$

Let $\tau = \inf\{t \geq 0 : B_t \in \{-a, b\}\}$, with $a, b > 0$ and $B_0 = 0$. Use the martingale B_t to compute $\mathbb{P}(B_\tau = b)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 95 – Brownian hitting probability via martingale

Solution: Detailed solution

1. At the stopping time, B_τ equals either b or $-a$.
2. Let $p = \mathbb{P}(B_\tau = b)$.
3. Then
$$\mathbb{E}[B_\tau] = bp + (-a)(1-p) = (a+b)p - a.$$
4. Optional stopping for the bounded interval exit gives $\mathbb{E}[B_\tau] = B_0 = 0$.
5. Solve:
$$(a+b)p - a = 0, \qquad p = \frac{a}{a+b}.$$

Exercise 96 – Brownian exponential martingale

Exam-style exercise: Brownian exponential martingale $\star\star\star\star$

Show that $M_t = \exp(\theta B_t - \frac{1}{2}\theta^2 t)$ has constant expectation one.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 96 – Brownian exponential martingale

Solution: Detailed solution

1. Since $B_t \sim N(0,t)$, its moment generating function is
$$\mathbb{E}[e^{\theta B_t}] = e^{\frac{1}{2}\theta^2 t}.$$
2. Therefore
$$\mathbb{E}[M_t] = e^{-\frac{1}{2}\theta^2 t}\mathbb{E}[e^{\theta B_t}].$$
3. Substitute the Gaussian moment generating function:
$$\mathbb{E}[M_t] = e^{-\frac{1}{2}\theta^2 t} e^{\frac{1}{2}\theta^2 t} = 1.$$
4. With independent increments one can also show the full martingale property.

Module 4 Exercise Pack

Poisson processes and counting dynamics

Exercise 97 – Poisson distribution

Exam-style exercise: Poisson distribution ⋆⋆

Let (N_t) be a Poisson process with rate λ. Compute $\mathbb{P}(N_t = k)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 97 – Poisson distribution

Solution: Detailed solution

1. By definition, N_t counts the number of arrivals up to time t.
2. For a Poisson process with rate λ, $N_t \sim \mathrm{Poisson}(\lambda t)$.
3. Therefore

$$\mathbb{P}(N_t = k) = e^{-\lambda t}\,\frac{(\lambda t)^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

Exercise 98 – Poisson increment

Exam-style exercise: Poisson increment $\star\star$

For $0 \leq s < t$, find the distribution of $N_t - N_s$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 98 – Poisson increment

Solution: Detailed solution

1. A Poisson process has stationary increments.
2. The increment $N_t - N_s$ counts arrivals in an interval of length $t - s$.
3. Hence
$$N_t - N_s \sim \text{Poisson}(\lambda(t - s)).$$
4. The distribution depends on the interval length, not on the starting time.

Exercise 99 – Poisson independent increments

Exam-style exercise: Poisson independent increments ⋆⋆

Are $N_3 - N_1$ and $N_5 - N_3$ independent for a Poisson process? Explain.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 99 – Poisson independent increments

Solution: Detailed solution

1. The first increment counts arrivals in $(1, 3]$.
2. The second counts arrivals in $(3, 5]$.
3. These intervals are disjoint.
4. A Poisson process has independent increments on disjoint intervals.
5. Therefore the two increments are independent.

Exercise 100 – Mean and variance of Poisson process

Exam-style exercise: Mean and variance of Poisson process $\star$

For a Poisson process with rate λ, compute $\mathbb{E}[N_t]$ and $\mathrm{Var}(N_t)$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 100 – Mean and variance of Poisson process

Solution: Detailed solution

1. Since $N_t \sim \text{Poisson}(\lambda t)$, its parameter is λt.
2. A Poisson random variable with parameter m has mean m and variance m.
3. Therefore

$$\mathbb{E}[N_t] = \lambda t, \qquad \text{Var}(N_t) = \lambda t.$$

Exercise 101 – Compensated Poisson martingale

Exam-style exercise: Compensated Poisson martingale ⋆⋆⋆

Show that $M_t = N_t - \lambda t$ is a martingale with respect to the natural filtration of the Poisson process.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 101 – Compensated Poisson martingale

Solution: Detailed solution

1. For $s < t$, write
$$N_t = N_s + (N_t - N_s).$$
2. The increment $N_t - N_s$ is independent of $\mathcal{F}_s$ and has mean $\lambda(t-s)$.
3. Thus
$$\mathbb{E}[N_t \mid \mathcal{F}_s] = N_s + \lambda(t-s).$$
4. Therefore
$$\mathbb{E}[N_t - \lambda t \mid \mathcal{F}_s] = N_s + \lambda(t-s) - \lambda t = N_s - \lambda s.$$
5. Hence $M_t = N_t - \lambda t$ is a martingale.

Exercise 102 – No arrival probability

Exam-style exercise: No arrival probability ⋆

For a Poisson process with rate λ, compute the probability of no arrivals in $[0, t]$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 102 – No arrival probability

Solution: Detailed solution

1. No arrivals means $N_t = 0$.
2. Since $N_t \sim \text{Poisson}(\lambda t)$,
$$\mathbb{P}(N_t = 0) = e^{-\lambda t} \frac{(\lambda t)^0}{0!} = e^{-\lambda t}.$$
3. Thus the no-arrival probability decays exponentially in time.

Exercise 103 – First arrival time

Exam-style exercise: First arrival time ⋆⋆

Let $T_1 = \inf\{t : N_t \geq 1\}$. Find $\mathbb{P}(T_1 > t)$ and identify the distribution of T_1.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 103 – First arrival time

Solution: Detailed solution

1. The event $\{T_1 > t\}$ means that no arrival has occurred by time t.
2. Hence
$$\mathbb{P}(T_1 > t) = \mathbb{P}(N_t = 0) = e^{-\lambda t}.$$
3. This is the survival function of an exponential random variable with rate λ.
4. Therefore $T_1 \sim \mathrm{Exp}(\lambda)$.

Exercise 104 – Conditional on total count

Exam-style exercise: Conditional on total count ⋆⋆⋆

For a Poisson process, compute $\mathbb{P}(N_s = k \mid N_t = n)$ for $0 < s < t$ and $0 \le k \le n$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 104 – Conditional on total count

Solution: Detailed solution

1. Conditional on $N_t = n$, the n arrival times are distributed like ordered independent uniform points on $[0, t]$.
2. The number falling in $[0, s]$ is binomial with success probability s/t.
3. Therefore

$$\mathbb{P}(N_s = k \mid N_t = n) = \binom{n}{k} \left(\frac{s}{t}\right)^k \left(1 - \frac{s}{t}\right)^{n-k}.$$

Exercise 105 – Superposition

Exam-style exercise: Superposition ⋆⋆⋆

Let $N^{(1)}$ and $N^{(2)}$ be independent Poisson processes with rates λ_1 and λ_2. What is the law of $N_t = N_t^{(1)} + N_t^{(2)}$?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 105 – Superposition

Solution: Detailed solution

1. At time t, $N_t^{(1)} \sim \text{Poisson}(\lambda_1 t)$ and $N_t^{(2)} \sim \text{Poisson}(\lambda_2 t)$.
2. The two variables are independent.
3. The sum of independent Poisson random variables is Poisson with parameter equal to the sum of parameters.
4. Hence
$$N_t \sim \text{Poisson}((\lambda_1 + \lambda_2)t).$$
5. The process-level result is that the superposition is a Poisson process with rate $\lambda_1 + \lambda_2$.

Exercise 106 – Thinning

Exam-style exercise: Thinning ⋆⋆⋆

Each arrival of a Poisson process with rate λ is independently kept with probability q. What is the rate of the kept process?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 106 – Thinning

Solution: Detailed solution

1. Conditional on $N_t = n$, the number of kept arrivals is $\mathrm{Bin}(n, q)$.
2. Mixing a binomial thinning with a Poisson count gives another Poisson count.
3. The new parameter over $[0, t]$ is $q\lambda t$.
4. Therefore the kept process is a Poisson process with rate $q\lambda$.

Exercise 107 – Poisson process is not stationary

Exam-style exercise: Poisson process is not stationary ⋆⋆

Is (N_t) stationary? Is it a process with stationary increments?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 107 – Poisson process is not stationary

Solution: Detailed solution

1. $N_t \sim \text{Poisson}(\lambda t)$, so its distribution changes with t.
2. Therefore (N_t) is not stationary.
3. But increments satisfy
$$N_{t+h} - N_t \sim \text{Poisson}(\lambda h),$$
which depends only on h.
4. Hence the process has stationary increments.

Exercise 108 – Jump times and counts

Exam-style exercise: Jump times and counts ⋆⋆

Let T_3 be the time of the third arrival of a Poisson process. Express $\{T_3 \leq t\}$ in terms of N_t.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 108 – Jump times and counts

Solution: Detailed solution

1. The third arrival occurs by time t if and only if at least three arrivals have occurred by time t.
2. Therefore
$$\{T_3 \leq t\} = \{N_t \geq 3\}.$$
3. This identity is often the easiest way to move between arrival times and counts.

Module 4 Exercise Pack

Integrated exam-style problems

Exercise 109 – Classify three processes

Exam-style exercise: Classify three processes ⋆⋆

Classify each process as stationary, having stationary increments, both or neither: white noise, symmetric random walk, Brownian motion.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 109 – Classify three processes

Solution: Detailed solution

1. White noise is stationary because its mean, variance and covariance structure do not change with time. Stationary increments are not its main defining property.
2. The symmetric random walk is not stationary because $\mathrm{Var}(S_n) = n$, but it has stationary increments.
3. Brownian motion is not stationary because $\mathrm{Var}(B_t) = t$, but it has stationary increments because $B_{t+h} - B_t \sim N(0, h)$.

Exercise 110 – Martingale or Markov?

Exam-style exercise: Martingale or Markov? ⋆⋆

Can a process be Markov but not a martingale? Give an example and justify it.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 110 – Martingale or Markov?

Solution: Detailed solution

1. Yes. A biased random walk is Markov because the next state depends on the present state through the next independent increment.
2. If $p \neq 1/2$, then
$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + (2p - 1) \neq S_n.$$
3. Therefore it is not a martingale.
4. Thus Markov and martingale are different properties.

Exercise 111 – Martingale but not independent increments

Exam-style exercise: Martingale but not independent increments $\star\star\star$

Let $M_n = Z\xi_n$, where Z and (ξ_n) are independent, Z takes values ± 1, and (ξ_n) is a symmetric random walk martingale difference sequence arranged so that $\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0$. Can a martingale fail to have independent increments?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 111 – Martingale but not independent increments

Solution: Detailed solution

1. Yes. The martingale property only requires zero conditional drift.
2. It does not require increments to be independent.
3. A common random factor such as Z can create dependence among increments while preserving conditional mean zero under an appropriate filtration.
4. Therefore independent increments imply strong structure, but martingales need not have independent increments.

Exercise 112 – Use the correct filtration

Exam-style exercise: Use the correct filtration $\star\star\star$

Let S_n be a symmetric random walk and define $\mathcal{G}_n = \{\emptyset, \Omega\}$ for all n. Is S_n a martingale with respect to $(\mathcal{G}_n)$?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 112 – Use the correct filtration

Solution: Detailed solution

1. A martingale must be adapted to the filtration.
2. Adaptation to $\mathcal{G}_n$ would require S_n to be $\mathcal{G}_n$-measurable.
3. A $\mathcal{G}_n$-measurable random variable must be constant.
4. But S_n is not constant for $n \geq 1$.
5. Therefore S_n is not a martingale with respect to the trivial filtration, even though it is a martingale with respect to its natural filtration.

Exercise 113 – Compare Brownian and Poisson martingales

Exam-style exercise: Compare Brownian and Poisson martingales ⋆⋆

Write one martingale naturally associated with Brownian motion and one with a Poisson process.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 113 – Compare Brownian and Poisson martingales

Solution: Detailed solution

1. For Brownian motion, B_t itself is a martingale.
2. Also $B_t^2 - t$ is a martingale.
3. For a Poisson process with rate λ, the compensated process

$$N_t - \lambda t$$

   is a martingale.
4. The common idea is compensation: subtract the predictable drift or accumulated variance.

Exercise 114 – Conditional expectation and Markov chains

Exam-style exercise: Conditional expectation and Markov chains ⋆ ⋆ ⋆

Let (X_n) be a Markov chain and let f be a bounded function. Show that $\mathbb{E}[f(X_{n+1}) \mid X_0, \ldots, X_n] = \mathbb{E}[f(X_{n+1}) \mid X_n]$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 114 – Conditional expectation and Markov chains

Solution: Detailed solution

1. The Markov property says that the conditional law of X_{n+1} given the whole past depends only on X_n.
2. Conditional expectation of $f(X_{n+1})$ depends only on this conditional law.
3. Therefore the whole past can be replaced by the present state:

$$\mathbb{E}[f(X_{n+1}) \mid X_0, \dots, X_n] = \mathbb{E}[f(X_{n+1}) \mid X_n].$$

4. If the chain has transition matrix P, the right-hand side is $(Pf)(X_n)$.

Exercise 115 – Transition operator

Exam-style exercise: Transition operator ⋆⋆⋆

For a Markov chain with transition matrix P and a function f on the state space, define $(Pf)(i)$ and interpret it.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 115 – Transition operator

Solution: Detailed solution

1. The transition operator acts by
$$(Pf)(i) = \sum_j p_{ij} f(j).$$
2. Starting from state i, the next state is j with probability p_{ij}.
3. Thus $(Pf)(i)$ is the conditional expectation of $f(X_{n+1})$ given $X_n = i$.
4. In formula:
$$\mathbb{E}[f(X_{n+1}) \mid X_n = i] = (Pf)(i).$$

Exercise 116 – Harmonic function and martingale

Exam-style exercise: Harmonic function and martingale $\star\star\star$

Let (X_n) be a Markov chain and suppose h satisfies $Ph = h$. Show that $h(X_n)$ is a martingale.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 116 – Harmonic function and martingale

Solution: Detailed solution

1. We need to compute $\mathbb{E}[h(X_{n+1}) \mid \mathcal{F}_n]$.
2. By the Markov property,
$$\mathbb{E}[h(X_{n+1}) \mid \mathcal{F}_n] = (Ph)(X_n).$$
3. Since $Ph = h$,
$$(Ph)(X_n) = h(X_n).$$
4. Therefore
$$\mathbb{E}[h(X_{n+1}) \mid \mathcal{F}_n] = h(X_n).$$
5. Hence $h(X_n)$ is a martingale.

Exercise 117 – Subharmonic function and submartingale

Exam-style exercise: Subharmonic function and submartingale $\star\star\star\star$

Let (X_n) be a Markov chain and suppose $Ph \geq h$. What can you say about $h(X_n)$?

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 117 – Subharmonic function and submartingale

Solution: Detailed solution

1. Compute with the Markov property:
$$\mathbb{E}[h(X_{n+1}) \mid \mathcal{F}_n] = (Ph)(X_n).$$
2. If $Ph \geq h$, then
$$(Ph)(X_n) \geq h(X_n).$$
3. Therefore
$$\mathbb{E}[h(X_{n+1}) \mid \mathcal{F}_n] \geq h(X_n).$$
4. Thus $h(X_n)$ is a submartingale, assuming integrability.

Exercise 118 – Brownian Markov property

Exam-style exercise: Brownian Markov property $\star\star\star$

Explain why Brownian motion is Markov using the decomposition $B_t = B_s + (B_t - B_s)$ for $s < t$.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 118 – Brownian Markov property

Solution: Detailed solution

1. Given $\mathcal{F}_s$, the value B_s is known.
2. The future increment $B_t - B_s$ is independent of $\mathcal{F}_s$.
3. Thus the conditional distribution of B_t given the entire past depends only on B_s.
4. More precisely, given $B_s = x$, B_t has law $x + N(0, t - s)$.
5. Hence Brownian motion is Markov.

Exercise 119 – Poisson Markov property

Exam-style exercise: Poisson Markov property ⋆⋆⋆

Explain why a Poisson process is Markov.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 119 – Poisson Markov property

Solution: Detailed solution

1. For $s < t$, write
$$N_t = N_s + (N_t - N_s).$$
2. The future increment $N_t - N_s$ is independent of the past $\mathcal{F}_s$.
3. Its distribution is $\text{Poisson}(\lambda(t-s))$.
4. Therefore, given the present count N_s, the future distribution of N_t is determined.
5. The past before s gives no additional information about future increments.

Exercise 120 – Final synthesis

Exam-style exercise: Final synthesis $\star\star\star$

A process has independent stationary increments and starts at zero. Name two examples from the module and one martingale associated with each.

Intuition: Before solving

Identify the object involved, write the relevant definition, then compute only after the notation is clear.

Solution 120 – Final synthesis

Solution: Detailed solution

1. Brownian motion has independent stationary increments and $B_0 = 0$.
2. A martingale associated with it is B_t, or $B_t^2 - t$.
3. A Poisson process has independent stationary increments and $N_0 = 0$.
4. A martingale associated with it is the compensated process $N_t - \lambda t$.
5. The common structure is: independent increments allow explicit conditional expectations.

Final checklist for Module 4 exam exercises

- Always specify the filtration before proving a martingale or stopping-time property.
- Equal marginal laws do not determine a stochastic process.
- Stationarity and stationary increments are different concepts.
- The Markov property concerns conditional laws given the present state.
- Martingales require adaptation, integrability and the conditional expectation identity.
- Optional stopping requires hypotheses; do not apply it mechanically.
- Brownian motion and Poisson processes are the two main continuous-time examples to master.